

DISASTER MANAGEMENT

BIOLOGICAL DISASTERS

MODULE - 9



DISASTER MANAGEMENT

DISASTERS ARISING FROM NATURAL HAZARDS

GEOPHYSICAL

- Earthquakes
- Volcano
- Tsunami



HYDROLOGICAL

- Avalanche
- Coastal Erosion
- Coastal Flood
- Hydrological flood
- Flash Flood
- Debris/mud flow



METEOROLOGICAL

- Cyclone
- Storm Surge
- Heat wave/ cold wave
- Heavy rains, Sand or dust storm



CLIMATOLOGICAL

- Droughts
- Extreme hot or cold conditions
- Forest fires
- Glacial lake outburst



BIOLOGICAL

- Epidemics – viral, bacterial, parasitic, fungal or prion infections
- Insect infestations
- Animal/Mass stampedes



BIOLOGICAL EMERGENCY AND BIOLOGICAL DISASTERS



- 1** Biological emergency is one caused due to
 - Natural outbreaks of epidemics
 - Intentional use of biological agents (viruses and microorganisms)
 - Toxins
- 2** The dissemination of such agents leads to harm human population, food crops and livestock to cause outbreaks of diseases.

- 3** This may happen through natural, accidental, or deliberate dispersal of such harmful agents into food, water, air, soil or into plants, crops, or livestock.
- 4** Apart from the natural transnational movement of the pathogenic organisms, their potential use as weapons of biological warfare and bio-terrorism has become far more important now than ever before.
- 5** Along with nuclear and chemical agents, many biological agents are now considered as capable of causing large-scale mortality and morbidity.



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BIOSAFETY LEVELS

BIOSAFETY LEVEL 1

No contaminant, unlikely to cause disease or to the environment, the pathogens that present minimal hazard. Ex – E.coli.

BIOSAFETY LEVEL 2

Contaminant, moderate risk associated with human disease/ environment hazard
Ex – Influenza, HIV

BIOSAFETY LEVEL 3

High contaminant, aerosol transmission, potentially lethal disease causing agent. Can be indigenous or exotic. Ex - tuberculosis

BIOSAFETY LEVEL 4

Maximum contaminant, exotic high risk agents with life threatening disease agents with high aerosol transmission. Ex- Ebola Virus



STEPS TO CONTROL BIOLOGICAL EMERGENCY



STEPS TO CONTROL BIOLOGICAL EMERGENCY



1 Handling exotic pathogens warrants **suitable infrastructure, notably, high containment laboratories** of bio-safety levels 3 and 4.

2 Recruitment of highly committed, dedicated and trained professionals.

3 **Biosafety laboratories** are required for the prompt diagnosis of the agents for effective management of biological disasters.

4 **Directorate of Plant Protection Quarantine & Storage** for minimizing crop losses through integrated pest management, plant quarantine, regulation of pesticides, locust warning & control and capacity building

5 Current system of surveillance and mechanism to control the outbreak of endemic diseases are through the National Programme for **Surveillance of Communicable Diseases**.



KEY WORDS TO LEARN ABOUT DISEASE OUTBREAKS

ENDEMIC

Indicates a disease or disaster risk that always affects a particular population.



EPIDEMIC

Indicates a disaster risk that affects many people in a given area or community.



PANDEMIC

Indicates a disaster risk that affects a much larger region, sometimes an entire continent or even the whole planet.





1

Epidemiological analysis of the surveillance data

2

Delivery of appropriate medical treatment and public health measures

3

Inventory control in the laboratories dealing with bacteria, viruses or toxins which can be a source of potential causative agents for biological disasters.

4

Installing HEPA filters in the ventilation systems of the air conditioning facilities will prevent infectious microbes from entering the air circulating inside critical buildings.

HOW TO
PREVENT
EPIDEMICS
DISASTER
RISK

8

Recognition and diagnosis by primary health care practitioners.

7

Communication of surveillance information to public health authorities

6

The post-exposure approach will include effective decontamination and safety

5

Buildings and Offices Protection of important buildings against biological agents wherever deemed necessary, can be done by preventing and restricting entry to authorised personnel only by proper screening.



1 The fast pace of development brings with it increasing frequency of various types of accidents as more and more people are involved in diverse economic activities.

2 While all these are matters of utmost concern, not all of them fall within the purview of the NDMP.

3 Certain specific agencies such as the Indian Coast Guard have the primary responsibility of addressing incidents of oil spills and ships in the coastal waters.

4 While the cases of fires in the built environment and forests are included in the plan, local authorities address them in accordance with the relevant emergency management systems.

5 The primary way to reduce risks is through mainstreaming risk reduction in development and governance.

6 As part of the overall DRR plan, systems for disaster preparedness and response are being strengthened at all levels



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INSECT INFESTATIONS IN INDIA



- 1 India is the only Bt cotton-growing country facing the problem of **pink bollworm infestation**.

- 2 All other Bt cotton-growing countries mainly grow open-pollinated cotton varieties rather than hybrids.

- 3 Even though Bollgard 2, or BG-2, Monsanto's second generation insecticidal technology for cotton, was supposed to protect crops against the pink bollworm, the pest has grown resistant to the toxins produced by this trait.

- 4 The pink bollworm grew resistant because India restricted itself to cultivating long-duration hybrids since the introduction of Bt cotton in 2002.

- 5 As a result, farmers now spend more on pesticides to control infestations.



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INSECT INFESTATIONS IN INDIA

- 1** Fall Armyworm (FAW), or *Spodoptera frugiperda*, is an insect that is native to tropical and subtropical regions of the Americas.
- 2** In its larva stage, it can cause significant damage to crops, if not well managed.
- 3** It prefers maize but can feed on more than 80 additional species of plants, including rice, sorghum, millet, sugarcane, vegetable crops, and cotton.
- 4** The pest was identified in Karnataka in May and has reached as far as West Bengal and Gujarat since then.
- 5** Climate change as well as rising trade and transportation between infected and non-infected regions as the reasons behind the invasion of Fall Armyworms.
- 6** The Directorate of Plant Protection, Quarantine & Storage work closely with the Ministry of agriculture and the State agricultural departments. **FAO Regional Representative for Asia** and the Pacific also looks into certain insect infestations.





EMERGENCIES ASSOCIATED WITH MASS GATHERINGS



- 1 Throughout the country, frequently, there are various kinds of events that attract crowds large and small, at varying types and styles of venue.
- 2 Inadequate planning can increase risks associated with insufficient or ineffective spectator management or service provision.
- 3 For the benefit of the state governments, local authorities and other agencies, NDMA has published a guideline on mass gatherings.
- 4 2019 has been an eventful year with all around massive surge in mass protests



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NDMA GUIDELINE OF A SIX MAJOR CAUSES AND TRIGGERS FOR CROWD DISASTERS



- 1 Structural** – The infrastructure, conditions and arrangements at the venue may not be adequate (collapse of barricades, fencing, temporary structures, insufficient exit, difficult terrain, slippery/muddy roads, etc.)

- 2 Fire/Electricity** – Risky practices involving fire and electricity ranging from makeshift facilities, shops, cooking, careless use of easily inflammable materials, non-availability or malfunctioning fire extinguishers, illegal electric connections, and many such possibilities
- 3 Crowd Control** – Crowds exceeding the capacity of the venue, poor management resulting in confusion and failure of all orderliness, not having enough emergency exits, inadequacy of systems to effectively communicate with the crowd and similar problems.



NDMA GUIDELINE OF A SIX MAJOR CAUSES AND TRIGGERS FOR CROWD DISASTERS



- 4 **Crowd Behavior** – There are numerous issues known to be associated with the behavior of crowds which is different from what is expected from an individual that tend to worsen emergency situations that may include unruly, irresponsible and angry responses.
- 5 **Security** – Under deployment of security personnel to regulate to control crowd, flaws in the planning of security arrangements
- 6 **Lack of Coordination between Stakeholders** – Significant coordination gap between agencies associated with the organizing of the event and authorities



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GREEN DISASTERS



1 In the wake of the recent protests by the urban youth and environmentalists, a new term of green disaster was coined in which the apathy of the government in protecting the environment is called out for.

2 One recent uprising is against the Maharashtra government regarding the cutting of Aarey Forests.

3 The Mumbai Metro Rail Corporation (MMRC) had approached the Tree Authority to seek permission for felling the trees to make way for a car shed which is a part of the Mumbai Metro project.

4 Aarey forests are known as the last 'lungs' of Mumbai which spurred citywide protests.

5 The Aarey forest is a habitat of its own. It is home to many animals and in a way is a vital part of the ecosystem.

6 The deprived animals can start entering residences if the forest is being cut.



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GREEN DISASTERS



- 7** The recent mass gatherings show from India to Iraq, Venezuela to Algeria, Haiti to Spain, Hong Kong to Colombia, Puerto Rico to Iran, **millions of people reached their breaking points.**

- 8** All over the globe massive protests have been carried out against surging global mean temperature.

- 9** Green disasters have awoken both the millennial and the younger generation.

- 10** 2019 was also the year of declaring 'Climate Emergencies' from cities to countries from UK, Ireland to New York City - as it became the world's largest city to declare climate emergency

Hundreds of people in Mumbai have been staging protests to show their disapproval of cutting down of around 2,700 trees in the Aarey forests.



CRITICAL ANALYSIS OF HOW INFRASTRUCTURE GROWTH GETS DERAILED BY GREEN ACTIVISM



- 1 India's ambition of sustaining its relatively high growth depends on one important factor: infrastructure.
- 2 The country, however, is plagued with a weak infrastructure incapable of meeting the needs of a growing economy and growing population.
- 3 India is striving to improve its manufacturing competitiveness at a time when manufacturing powerhouse China is shifting toward consumption-led growth.

- 4 Successful infrastructure development can provide a boost to many sectors, including steel, cement, auto, real estate, and others.
- 5 In India, key urban centers and manufacturing hubs also often struggle with capacity constraints, which directly affects competitiveness.
- 6 lower labor costs alone aren't enough to bolster India's manufacturing competitiveness. It will be sustainable only with adequate national infrastructure, which currently is lacking.
- 7 We have to strike a fine balance between destruction and development where protection of the environment does not bring down the opportunities of growth.





RELATED QUESTIONS THAT CAN BE ANSWERED AFTER READING THE MODULE



- 1** What are the endemic, epidemic and pandemic disease disaster risks? How can such biological emergencies be controlled for developing resilience in communities?
- 2** In the wake of the recent Economic Development vs. Environment debate reignited through Aarey Forest, how critical is the balance between sustainability and infrastructure.
- 3** The Fall Army Worm is a polyphagous (ability to feed on different kinds of food) caterpillar that has been on news due to maize crop failing. How these events can be tackled in the future to secure the food crops
- 4** Discuss the NDMA guidelines for causes and triggers for crowd aggressions and mass movements. Is today's society vulnerable to it?



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